LOCAL SELECTION ACTIVITY ·

From Curiosity to Concept

Team Selection Round

24 teams. 15 places. Choose one challenge.

Organizer	**NOVA for Astronomy and Space Science**
Event location	**A. M. Qattan Foundation**
Issue date	**9 Nov, 2026**
Submission deadline	**10 Nov, 2026 14:00 pm**
Submission link	Click Here to submit your work

- Note: This selection round is organized locally. It has not been issued, reviewed or endorsed by NASA or the NASA Space Apps program.

SECTION B

Welcome and overview

Thank you for applying. We received more applications than we have places, so we are running a short selection round. **Fifteen of the 24 applicant teams will join our event.** This round is not the hackathon itself. It is a chance for your team to show how it thinks, researches, decides and communicates.

Purpose	To see how teams solve problems, reason with evidence, create original ideas and work together, using a small, manageable task.
Your choice	Each team chooses **ONE** of two challenges (A or B) and submits one solution.
Same yardstick	The **same evaluation framework** applies to both challenges: one rubric, 100 points.
No prototype needed	A clear, well-supported concept can score highly **without a working prototype**.
Time	You have **24 HOURS** from **8 Nov 14:00** to **9 Nov 14:00**. We expect about 4–8 hours of active team effort.
Results	Selected teams will be announced on **10 Nov**.

WHAT WE ARE LOOKING FOR

- **Curiosity and a clear problem focus:** you understand who is affected and why it matters.
- **Evidence-based reasoning:** you use sources well and are honest about what is assumption.
- **Original, well-justified ideas:** you can explain why you chose your approach over others.
- **Realism:** you show what is feasible, name limits and plan a simple way to test your idea.
- **Clear communication and real teamwork:** others can follow your thinking, and every member's contribution is visible.



Good to know

Ideas that are simple, clear and well reasoned will do better than complex ideas that are hard to explain. Expensive equipment, advanced programming and polished visuals do not earn points on their own.

SECTION C

General instructions

Original work and attribution	The submission must be your team's own work. Credit every source, image, dataset and idea you use. Do not copy another team's solution or a previous hackathon project.
External resources, code, images and datasets	You may use freely accessible tools, code, images and public datasets, if you are allowed to use them and you cite them. Say clearly what you created and what you reused.
AI use (proposed policy)	You may use AI tools responsibly, for example to brainstorm, summarize, check language or help with code. You must **disclose** which tools you used and for what, and your team must **verify every claim, number, source and output**. AI-generated references must be checked to exist. AI use earns no points. Unchecked or fabricated content will lower your scores, and missing or inaccurate disclosure may affect eligibility.
Equal treatment of formats	Storyboards, diagrams, wireframes, models, visualizations and prototypes are all assessed on the same rubric. We assess what your work shows, not how technical or polished it is.
Files and naming	Submit one PDF report or video, named **TEAMNAME_Challenge-A_Report.pdf** or **TEAMNAME_Challenge-A_Video.mp4** (use B for Challenge B; no spaces in the team name). Upload at **the submission form**
Language	Report or video must be in English.
Deadline and updates	Submit by 10 Nov 14:00 pm . If you upload more than once, only the latest version received before the deadline is reviewed.
Questions	Write to **khalilmahajneh0@gmail.com**. Answers to general questions may be shared with all teams to keep things fair.

SECTION D

Standard submission requirements

Both challenges require the concept report.

Item	Requirement
Concept report	One **PDF**, maximum **four pages** (text no smaller than 10 pt), including visuals and references.
Optional team video	One short video, maximum **three minutes**, explaining the problem, your proposed solution, major design decisions and each member's contribution. Team members can appear on camera, narrate over slides or use captions.
Optional prototype link	You may add a link to a prototype. It is **never required** and earns **no automatic bonus points**. We assess what it demonstrates, not how sophisticated it is.

REQUIRED REPORT SECTIONS

#	Section	What to include
1	Team identification and selected challenge	Team name, challenge (A or B), member names.
2	Problem and intended users	The specific problem you chose to solve and who it is for.
3	Proposed solution and how it works	Your idea and how its parts fit together.

#	Section	What to include
4	Supporting science, research or data	The evidence behind your main choices.
5	At least one meaningful visual	A diagram, wireframe, chart or storyboard that explains your idea.
6	Feasibility, limitations and validation plan	What is realistic, what could go wrong, and a simple way to test it.
7	Contributions, references and AI-use disclosure	Who did what; sources; AI tools used and how you verified outputs (write “none” if none); any reused work.

How we judge communication

Video production quality and English accent are **not** assessment criteria. We look for understandable communication and sound reasoning.

SECTION E

Challenge briefs

Choose **one** challenge. Both ask for a concept-level solution, use the same submission package and are judged on the same rubric. Original NASA Space Apps challenge titles and sources are shown to give credit; the tasks below are our local adaptations.

A

CHALLENGE A · ASTROBIOLOGY AND OCEAN WORLDS

Life Without Sunlight: Design a Chemosynthetic Ocean World

Original NASA Space Apps challenge	**Beyond Sunlight: An Aquatic Chemosynthetic World** 2024 NASA International Space Apps Challenge (held 5–6 October 2024)
Official sources	2024 challenge listing: https://www.spaceappschallenge.org/nasa-space-apps-2024/challenges/ Space Apps announcement of the 2024 challenges: https://www.spaceappschallenge.org/blog/ignite-your-ideas-with-a-challenge
How we adapted it	The original challenge invites teams to design an ocean world, and its ecosystem, where chemosynthesis rather than photosynthesis sustains life. Our version narrows this to **one habitat zone**, one energy pathway and a small food web, and asks teams to justify each design choice with cited science. No coding or simulation is needed.

BACKGROUND AND REAL-WORLD RELEVANCE

Most life we know depends on sunlight. In Earth's deep ocean, however, communities around hydrothermal vents and cold seeps are powered by **chemosynthesis**: microbes use chemical energy, such as from hydrogen sulfide or methane, to build the base of a food web.

Ocean worlds beyond Earth make this idea more than a curiosity. NASA's Cassini mission found evidence of hydrogen entering the ocean of Saturn's moon Enceladus through hydrothermal activity, a possible chemical energy source for life. "Habitable" does not mean "inhabited": this challenge is about reasoning carefully about what could be possible.

PROBLEM STATEMENT

> Students and the public find it hard to imagine how an ecosystem could run without sunlight. **Design a fictional but scientifically reasoned ocean world, focused on one habitat zone, in which chemosynthesis is the base of the food web, and explain it so a non-specialist can follow your reasoning.**

YOUR TASK

1. Define your habitat zone: where it is in your ocean world and its key conditions (for example depth or pressure, temperature range, main chemicals).
2. Identify the energy source and show how energy passes from it through microbes to at least two further levels of life.
3. Justify each major design choice with real science or observations, and say what is assumption rather than evidence.
4. Explain the limits of your idea and how scientists could test whether such a zone is plausible.

INTENDED USERS OR BENEFICIARIES

Students and members of the public learning about astrobiology, and the educators or science communicators who would use your concept to teach it. Scientists exploring where life might exist beyond Earth are a secondary audience.

FIVE GUIDING QUESTIONS

1. What process supplies energy in your world, and why would that supply last long enough for an ecosystem to depend on it?
2. Which chemical reactions could microbes use, and where on Earth or in our solar system is there evidence of something similar?
3. How do conditions such as pressure, temperature, salinity and darkness shape what organisms need in order to survive?
4. How does energy move from its source to the largest organism in your ecosystem, and what is lost along the way?

5. What would happen if the energy supply stopped, and what observation could show that your world is plausible?

MINIMUM SOLUTION REQUIREMENTS

- One named habitat zone with at least four stated conditions, including the energy source.
- An energy-flow diagram with at least three levels of life.
- At least three design decisions, each linked to a cited real-world fact.
- At least two stated limitations and a simple validation plan.
- At least four cited sources, two of them from NASA, NOAA or another space or science agency, or peer-reviewed.

SCOPE AND CONSTRAINTS

- **In scope:** fictional organisms and worlds that obey physics and chemistry you can cite; one zone, not a whole planet.
- **Out of scope:** claiming that life exists on a real world; unexplained or “magic” energy sources; detailed simulations or invented data presented as real.

ACCEPTABLE SOLUTION FORMATS

Annotated cross-section diagram, illustrated food web or energy map, storyboard, simple energy-budget model or table, short interactive page, or a photographed physical model. All formats are scored equally.

STARTING RESOURCES (ALL FREE TO ACCESS)

Resource	How it can help
NOAA Ocean Exploration: Chemosynthesis fact sheet Chemosynthesis Fact Sheet	Explains how vent and seep microbes turn chemicals into food, giving you real reactions to build on.
NOAA: Chemosynthesis and Hydrothermal Vent Life https://oceanexplorer.noaa.gov/edu/learning/player/lesson05/l5text.htm	Describes vent ecosystems and how fragile they are, which helps you think about energy supply and stability.
NOAA: Life Beyond Light poster (vents and seeps) https://oceanexplorer.noaa.gov/edu/materials/LifeBeyondLight-poster.pdf	Shows what lives where around vents and seeps, useful for building a realistic food web.
NASA: Missions provide new insights into “ocean worlds” https://science.nasa.gov/missions/cassini/nasa-missions-provide-new-insights-into-ocean-worlds-in-our-solar-system-2/	Links hydrothermal activity and hydrogen to the possible habitability of Enceladus and Europa.

CHALLENGE-SPECIFIC EVIDENCE EXPECTED

- Your energy-flow diagram, labeled with the energy source and each level of life.
- A short table: design choice, real-world evidence, source.
- A clear statement of assumptions and limitations, and your validation idea.

COMPLETE-SUBMISSION CHECKLIST

- ☐ Habitat zone and four conditions stated
- ☐ Energy source named and explained
- ☐ Energy-flow diagram with 3+ levels
- ☐ 3+ design choices linked to cited facts
- ☐ Assumptions and 2+ limitations stated
- ☐ Validation plan included
- ☐ 4+ sources cited (2+ agency or peer-reviewed)
- ☐ Standard report and video completed

B

CHALLENGE B · EARTH OBSERVATION AND AGRICULTURE

Water Wise: A Satellite-Informed Decision Concept for Farmers

BACKGROUND AND REAL-WORLD RELEVANCE

Farmers face water-related risks such as drought, flooding, or both in the same season. These affect crop health, income and food security. NASA satellites observe conditions that matter to these decisions, including rainfall (GPM), soil moisture (SMAP) and groundwater change (GRACE-FO).

The data are free to use, but turning satellite observations into guidance a farmer can act on is difficult. Where the data come from, how current they are and how detailed they are all affect whether they are useful. This challenge is about making that connection clearly and honestly.

PROBLEM STATEMENT

> Farmers make time-sensitive water decisions with limited information, and free satellite data are rarely presented in a form they can use. **Choose one farmer and one water-related decision, then design a concept that turns relevant NASA Earth observations into clear, actionable guidance, and explain its limits.**

YOUR TASK

1. Define your farmer (who, where, what type of farm) and the one water-related decision they face.
2. Identify which kinds of satellite observation could inform that decision, and what each one can and cannot tell the farmer.
3. Design how the guidance reaches the farmer and what they would do with it.
4. Explain the limits of your idea and how you would test it with farmers or local information.

INTENDED USERS OR BENEFICIARIES

Farmers of the type you choose, such as smallholder or commercial growers, and the people who support them, such as agricultural advisers or cooperatives. Describe your users specifically and explain why you chose them.

FIVE GUIDING QUESTIONS

1. Who exactly is your farmer, what water-related decision do they face and when, and what does a wrong decision cost?
2. What information does the farmer already use, and what is missing?
3. Which kinds of NASA observations are relevant to this decision, and how often are they updated and how detailed are they?
4. How will the guidance reach the farmer, considering device, language, connectivity and trust, and in what form?
5. How could you check that the guidance is correct and useful before building anything large?

MINIMUM SOLUTION REQUIREMENTS

- A farmer profile, region and one clearly stated decision.
- A data-to-decision map linking at least two observation types to the decision, noting update frequency and detail from a cited source.
- One real NASA observation example (a screenshot or cited figure with date, place and what it shows).
- A user-flow visual such as a wireframe or storyboard.
- At least two stated limitations and a simple validation plan.
- At least four cited sources, two of them from NASA or another space agency, or peer-reviewed.

SCOPE AND CONSTRAINTS

- **In scope:** one decision for one farmer type; browsing existing NASA imagery or reading NASA pages for your example.
- **Out of scope:** downloading or processing datasets; building a working app; claiming accuracy you have not tested; professional agronomic advice; any suggestion that NASA endorses your concept.

ACCEPTABLE SOLUTION FORMATS

Wireframe or mock-up, storyboard of a farmer's day, annotated data-to-decision diagram, simple decision flowchart or table, short interactive page, or a basic prototype. All formats are scored equally.

STARTING RESOURCES (ALL FREE TO ACCESS)

Resource	How it can help
NASA: From Seed to Market, NASA Brings Food to the Table https://www.nasa.gov/missions/gpm/from-seed-to-market-nasa-brings-food-to-the-table	Gives an overview of how rainfall, soil-moisture and groundwater observations relate to farming decisions.
NASA GPM: Soil moisture applications https://gpm.nasa.gov/applications/soil-moisture	Explains how rainfall data and soil-moisture observations connect, useful for your data-to-decision map.
NASA Worldview https://worldview.earthdata.nasa.gov	Lets you browse daily satellite imagery in a web browser, so you can find a real example without coding.
NASA Science: SMAP mission (Soil Moisture Active Passive) https://science.nasa.gov/mission/smap	Summarizes what SMAP measures (soil moisture and freeze/thaw state), helping you name the right observation type for your decision.

CHALLENGE-SPECIFIC EVIDENCE EXPECTED

- Your data-to-decision map, with each observation type and its update frequency or detail.
- One real NASA observation example, with date, location and what it shows.
- A clear statement of limitations (for example detail, delay, cloud cover or access) and your validation idea.

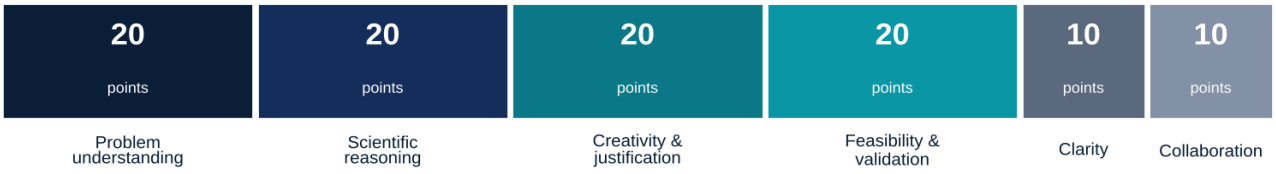
COMPLETE-SUBMISSION CHECKLIST

- ☐ Farmer, region and one decision stated
- ☐ 2+ observation types linked to the decision
- ☐ One real NASA example with date and place
- ☐ User-flow visual included
- ☐ 2+ limitations stated, plus validation plan
- ☐ 4+ sources cited (2+ agency or peer-reviewed)
- ☐ Update frequency or detail noted from a source
- ☐ Standard report and video completed

SECTION F

Evaluation rubric

One common rubric, **100 points**, is used for both challenges. Each submission is scored independently by two judges and the scores are averaged. We do not give points simply for using AI, writing code, building expensive prototypes or producing visually elaborate presentations.



Problem understanding and relevance 20 points		
Excellent	**17–20**	The problem and users are precisely defined and narrowed to a manageable focus. The real-world or scientific relevance is clear and the challenge's intent is faithfully followed.
Strong	**12–16**	The problem and users are clear with minor vagueness. Relevance is explained and the focus is mostly appropriate.
Developing	**6–11**	The problem is broad or generic. Users or relevance are only partly explained, or the work drifts from the challenge.
Limited	**0–5**	The problem is unclear or off-topic, only restates the challenge, or users are not identified.

Scientific reasoning and use of evidence 20 points		
Excellent	**17–20**	Key claims are explained with correct science or data. Sources are accurate, meet the minimum and are tied to specific decisions. Evidence and assumptions are clearly separated.
Strong	**12–16**	Reasoning is mostly correct and sources are linked to most decisions. There are minor gaps or unflagged assumptions.
Developing	**6–11**	Some correct facts appear but are weakly linked to the design. Sources are listed but little used.
Limited	**0–5**	Claims are unsupported or incorrect, or sources are missing or cannot be verified.

Creativity and justification of the approach 20 points		
Excellent	**17–20**	The approach is original or a thoughtful combination of ideas. The team explains why it chose this approach and compares at least one alternative.
Strong	**12–16**	There is a fresh element and the choice is reasonably explained.
Developing	**6–11**	The idea is familiar and there is little justification for choosing it.
Limited	**0–5**	The idea is generic or copied, with no justification.

Feasibility and proposed validation 20 points

Excellent	**17–20**	The concept is realistic within the stated constraints. Limits and risks are named honestly, and the validation plan is specific (what, who, how, what counts as success) and achievable.
Strong	**12–16**	The concept is realistic and limitations are noted. The validation plan exists but is partly general.
Developing	**6–11**	Feasibility is asserted rather than shown. Limits are vague and validation is generic.
Limited	**0–5**	The concept is unrealistic, or limits and validation are missing.

Clarity of communication 10 points		
Excellent	**9–10**	Report and video are easy to follow and logically ordered. Visuals explain the idea and terms are defined.
Strong	**7–8**	Mostly clear, with minor lapses in order or explanation.
Developing	**4–6**	Hard to follow in places, or visuals are decorative rather than explanatory.
Limited	**0–3**	Key parts are confusing or missing.

Team collaboration and documented contributions 10 points		
Excellent	**9–10**	Every member's contribution is specific and credible, and there is evidence of shared decisions or iteration.
Strong	**7–8**	Contributions are listed for all members with some detail.
Developing	**4–6**	Contributions are generic ("everyone helped") or very uneven.
Limited	**0–3**	Contributions are missing or unclear.

Totals and ties

Total = 20 + 20 + 20 + 20 + 10 + 10 = **100 points**. The 15 highest-scoring eligible teams across both challenges are selected; there is **no quota** for either challenge. Ties are broken by higher Scientific reasoning, then Problem understanding, then Feasibility, then an independent review.